ASSIGNMENT

Innovation Management & Intellectual Property Rights

Course Outcomes: CO1 | CO2 | CO3

Bloom's Taxonomy: BT2 – Understand | BT3 – Apply

Total Marks: 40

Subject	Innovation & Entrepreneurship
Assignment Type	Written Assignment
Total Questions	3
Total Marks	40
Date of Submission	____________________
Student Name	____________________
Roll Number	____________________

Assignment Overview

This assignment comprises three questions covering the fundamentals of Innovation Management and Intellectual Property Rights (IPR). Students are expected to demonstrate conceptual understanding (BT2) and practical application (BT3) aligned with the prescribed Course Outcomes.

Q.No	Topic	Marks	BT	CO
1	Current & Emerging Trends in Innovation	16	BT2	CO1
2	Importance of IPR – Patents, Copyrights & GI	16	BT2	CO2
3	Legal & Regulatory Process for IPR Registration	8	BT3	CO3
	TOTAL	**40**		

Evaluation Rubrics

Criteria	Excellent (9–10)	Good (7–8)	Average (5–6)	Poor (0–4)	Max
Conceptual Understanding	Complete & accurate grasp of all concepts	Good understanding with minor gaps	Basic understanding; limited explanation	Very limited understanding	**10**
Content Coverage	All key points explained with clarity	Most points covered	Some important points missing	Very few points covered	**10**
Application & Examples	Relevant examples & applications clearly explained	Some examples provided	Few examples with limited relevance	No relevant examples	**10**
Organisation & Presentation	Very well structured & clearly presented	Mostly organised	Some structure but lacks clarity	Poorly organised	**10**

Question 1 | 16 Marks Marks | BT2 – Understand | CO1

Outline the Current and Emerging Trends in Innovation

1.1 Introduction to Innovation

Innovation is the process of translating an idea or invention into a product, service, or methodology that creates value and for which customers will pay. It is the engine of economic growth, competitive advantage, and societal progress. Joseph Schumpeter's concept of **'Creative Destruction'** captures the essence: new innovations displace outdated technologies, business models, and industries, continuously reshaping the economic landscape.

> **Definition (OECD Oslo Manual):** Innovation is the implementation of a new or significantly improved product (good or service), process, new marketing method, or new organisational method in business practices, workplace organisation, or external relations.

1.2 Types of Innovation

- **Product Innovation**: Creation of new or improved goods/services — e.g., iPhone, Electric Vehicles.
- **Process Innovation**: New or improved production/delivery methods — e.g., Toyota Production System, Agile software development.
- **Business Model Innovation**: Redesigning how value is created, delivered and captured — e.g., Netflix subscription model, Uber ride-hailing.
- **Organisational Innovation**: New methods in workplace organisation or business practices — e.g., flat hierarchies, remote-first work culture.
- **Marketing Innovation**: Novel marketing strategies — e.g., influencer marketing, personalised AI-driven advertisements.

1.3 Major Current Trends in Innovation

Digital Transformation

Organisations across every sector are embedding digital technologies into their core operations. Cloud computing, big data analytics, and mobile-first strategies have become standard. Companies like Amazon and Flipkart have transformed retail by leveraging real-time data, automated warehouses, and predictive logistics. The COVID-19 pandemic accelerated this trend, compressing five years of digital adoption into eighteen months across industries.

Artificial Intelligence & Machine Learning

AI and ML are reshaping decision-making, customer experience, and operational efficiency. ChatGPT and large language models (LLMs) have democratised access to sophisticated AI. In healthcare, AI-driven diagnostics (e.g., Google DeepMind's AlphaFold for protein structure prediction) are transforming drug discovery. In finance, algorithmic trading and fraud detection rely entirely on ML models.

Internet of Things (IoT)

Billions of devices are now interconnected, generating continuous streams of data. Smart cities, connected factories (Industry 4.0), precision agriculture, and wearable health monitors exemplify IoT's breadth. India's Smart Cities Mission is a policy-level initiative that directly harnesses IoT infrastructure for urban governance.

Sustainable & Green Innovation

Climate change imperatives are driving massive investment in clean energy, circular economy models, and carbon-neutral manufacturing. Tesla's electric vehicles, ISRO's use of green propellants, and startups like Ather Energy in India illustrate how sustainability is becoming central to product strategy rather than an afterthought.

Open Innovation & Ecosystem Collaboration

Henry Chesbrough's open innovation paradigm sees firms sourcing ideas externally from startups, universities, and customers. Hackathons, accelerators (e.g., IIT Madras's Research Park), and API-driven platforms (e.g., Aadhaar's open APIs) exemplify collaborative innovation at scale.

1.4 Emerging Technological Innovations

Generative AI & Foundation Models

Models such as GPT-4, Gemini, and Claude can generate text, images, code, and audio with near-human fluency. Businesses are deploying these for content creation, coding assistance, legal document drafting, and customer service automation, fundamentally altering knowledge work.

Quantum Computing

Quantum computers exploit superposition and entanglement to solve problems intractable for classical machines. IBM's 1,000-qubit Condor processor and Google's Sycamore represent milestones. Quantum supremacy promises breakthroughs in cryptography, materials science, and pharmaceutical modelling.

Biotechnology & Genomics

CRISPR-Cas9 gene editing, mRNA vaccines (COVID-19 vaccines by Pfizer-BioNTech), and synthetic biology are redefining medicine and agriculture. India's bioeconomy, valued at over USD 150 billion by 2025, is a major emerging frontier.

Extended Reality (XR) – AR/VR/MR

Metaverse platforms (Meta's Horizon, Apple Vision Pro) and industrial AR solutions (e.g., PTC ThingWorx) are blurring physical and digital boundaries. Education, surgery simulation, and remote maintenance are key application domains.

Blockchain & Decentralised Technologies

Beyond cryptocurrency, blockchain ensures supply chain provenance, transparent voting systems, and secure digital identity. India's National Blockchain Framework is exploring government-use-case pilots in land records and credential verification.

5G & Advanced Connectivity

5G networks provide ultra-low latency and massive device connectivity, enabling autonomous vehicles, remote surgery, and smart manufacturing. The Telecom Regulatory Authority of India (TRAI) is spearheading 5G ecosystem development to catalyse Industry 4.0 adoption.

1.5 Impact on Business and Society

Domain	Impact of Innovation	Example
Business Models	Disruption of traditional revenue streams; emergence of platform economy	Uber vs Traditional Taxis
Employment	Automation displaces routine jobs; new roles in AI, data science, robotics emerge	Amazon warehouses vs Retail clerks
Healthcare	Personalised medicine, telemedicine, AI diagnostics improve outcomes	Practo, Apollo 24|7
Education	E-learning, adaptive platforms, and AI tutors democratise quality education	NPTEL, Coursera, Khan Academy
Environment	Clean tech reduces emissions; circular economy minimises waste	Waaree Solar, ReNew Power
Governance	Digital governance, e-services, and data-driven policy improve public delivery	DigiLocker, UPI, CoWIN

Conclusion: *Innovation is no longer a luxury — it is a survival imperative. As disruptive technologies converge, organisations that embrace continuous innovation will define the next decade of economic and social progress. India's aspirations under initiatives such as Make in India, Startup India, and the National Innovation and Startup Policy underscore the strategic importance of fostering an innovation ecosystem.*

Question 2 | 16 Marks Marks | BT2 – Understand | CO2

Importance of Intellectual Property Rights (IPR) — Patents, Copyrights & Geographical Indications

2.1 Introduction to Intellectual Property Rights

Intellectual Property Rights (IPR) are legal entitlements granted by a sovereign state to creators and inventors, conferring exclusive rights to use, reproduce, and commercially exploit their intellectual creations for a defined period. The World Intellectual Property Organization (WIPO) and the Agreement on Trade-Related Aspects of Intellectual Property Rights (TRIPS) form the international framework governing IPR. In India, IPR policy is overseen by the Office of the Controller General of Patents, Designs & Trade Marks (CGPDTM) under the Department for Promotion of Industry and Internal Trade (DPIIT).

> **Key Principle:** IPR balances two competing interests — rewarding inventors with temporary monopoly rights to incentivise innovation, while ensuring eventual public access to knowledge once the protection period expires.

2.2 Patents

A **patent** is an exclusive right granted for an invention — a product or process that provides a new way of doing something or offers a new technical solution to a problem. In India, patents are governed by the *Patents Act, 1970* (amended in 2005), and grant protection for **20 years** from the date of filing.

Criteria for Patentability:

- **Novelty** — the invention must be new and not disclosed in prior art.
- **Inventive Step** — must not be obvious to a person skilled in the relevant field.
- **Industrial Applicability** — must be capable of being made or used in industry.

Example:

Novartis vs Union of India (2013) — The Supreme Court of India rejected Novartis's patent application for Gleevec (imatinib mesylate), a cancer drug, ruling that incremental modifications to known compounds do not meet India's stringent inventive-step requirement (Section 3(d) of the Patents Act). This landmark judgment balanced pharma IPR with public health access, demonstrating how patent law shapes drug pricing and accessibility.

Another example is **ISRO's patent portfolio** covering indigenous satellite communication and remote-sensing technologies, which protects national technological sovereignty while enabling international licensing revenues.

2.3 Copyrights

Copyright protects original literary, artistic, dramatic, musical, and software works. In India, copyright is governed by the *Copyright Act, 1957*. Protection arises **automatically upon creation** — registration is optional but advantageous as legal evidence. The duration is the **lifetime of the author plus 60 years**.

Categories Protected by Copyright:

- **Literary Works**: Novels, poems, software code, academic papers — e.g., J.K. Rowling's Harry Potter series.

• **Musical Works**: Compositions and lyrics — e.g., A.R. Rahman's film soundtracks are copyrighted separately from film copyrights.

• **Cinematographic Films**: Bollywood films — e.g., Yash Raj Films holds copyright over its entire film catalogue.

• **Computer Programs**: Software source code — e.g., Microsoft Windows and Adobe Photoshop source code.

• **Broadcast Reproduction**: Live streaming rights — e.g., BCCI's exclusive broadcast rights for IPL matches.

Significance: Copyright enables creators to monetise their work through licensing, royalties, and assignments. The Indian music industry's transition to streaming (Spotify, JioSaavn) is entirely underpinned by copyright licensing agreements with labels such as T-Series, Sony Music India, and Tips Films.

2.4 Geographical Indications (GI Tags)

A **Geographical Indication (GI)** is a sign used on products that have a specific geographical origin and possess qualities or a reputation due to that origin. India's GI regime is governed by the *Geographical Indications of Goods (Registration and Protection) Act, 1999*, and protection lasts for **10 years, renewable indefinitely**.

GI Product	State/Region	Significance
Darjeeling Tea	West Bengal	First Indian product to receive GI tag (2004); global premium bran
Basmati Rice	Punjab, Haryana, UP & others	Protected against foreign imitation; major export commodity.
Kanchipuram Silk Saree	Tamil Nadu	Protects traditional weavers; ensures authenticity of premium sare
Channapatna Toys	Karnataka	Preserves traditional lacquer craft; supports rural artisan communi
Alphonso Mango	Ratnagiri, Maharashtra	Prevents mislabelling; ensures premium pricing for farmers.
Tirupati Laddu	Andhra Pradesh	Prevents counterfeit religious offerings; protects temple's brand ec

2.5 Comparative Overview of IPR Forms

Aspect	Patent	Copyright	Geographical Indication
Governing Act	Patents Act, 1970	Copyright Act, 1957	GI Act, 1999
Subject Matter	Inventions & processes	Creative & literary works	Goods linked to geography
Duration	20 years	Life + 60 years	10 years (renewable)
Registration	Mandatory	Optional (automatic)	Mandatory
Exclusive Right	Manufacture, use, sell	Reproduce, distribute	Use of the GI tag
Key Body (India)	Patent Office, Chennai/Mumbai/Delhi/Kolkata	Copyright Office, New Delhi	GI Registry, Chennai

Conclusion: *IPR is the legal bedrock of the knowledge economy. By protecting inventors, creators, and regional producers, the IPR framework stimulates R&D investment, preserves cultural heritage, and ensures fair competition. India's rapidly growing patent filings (+15.73% in FY2023), GI registrations exceeding 600 products, and a vibrant creative industry collectively underscore the economic centrality of a robust IPR regime.*

Question 3 | 8 Marks Marks | BT3 – Apply | CO3

Legal & Regulatory Process for IPR Registration — Patent Registration in India

3.1 Overview

Among the various forms of IPR, the **patent registration process** is selected here for its complexity and practical relevance to technology entrepreneurs. The process is governed by the *Patents Act, 1970* and *Patent Rules, 2003*, and administered by the Indian Patent Office (IPO) with offices in Chennai, Mumbai, Kolkata, and New Delhi.

3.2 Step-by-Step Patent Registration Process

Step 1	**Idea Documentation & Patentability Search** Document the invention in detail — technical drawings, working principle, and claims. Conduct a prior-art search on databases such as **InPASS (Indian Patent Advanced Search System)**, USPTO, and Espacenet to verify novelty. This avoids costly rejection and helps refine claims.
Step 2	**Determine Applicant Eligibility** The applicant can be the inventor, assignee, or legal representative. For company-owned inventions, IP assignment agreements must be in place. NRIs and foreign entities can also file in India under TRIPS obligations.
Step 3	**File Provisional or Complete Application** **Provisional Application (Form 2)** — Establishes priority date; complete specification must follow within 12 months. **Complete Specification** — Contains full claims, abstract, description, and drawings. Filing fees: INR 1,600 (individual) to INR 8,000 (large entity).
Step 4	**Publication of Application** The application is published in the *Official Patent Journal* **18 months** after the priority date (automatic), or earlier upon request (Form 9, for a fee). Publication creates 'provisional protection' — the applicant can claim damages from the published date.
Step 5	**Request for Examination (RFE)** The applicant must file Form 18 (RFE) within **48 months** of the priority date. Without RFE, the application is treated as withdrawn. A Patent Examiner then conducts a substantive examination on novelty, inventive step, and industrial applicability.
Step 6	**First Examination Report (FER) & Response** The IPO issues a FER raising objections. The applicant must respond within **6 months** (extendable by 3 months). Multiple rounds of correspondence (prosecution) may follow to resolve objections on patentability, claim scope, or formal requirements.
Step 7	**Grant of Patent** Once all objections are resolved, the patent is **granted and published** in the Patent Journal. A Letters Patent certificate is issued. The patent is valid for **20 years** from the filing date, subject to annual renewal fees.

Step 8

Post-Grant Opposition & Maintenance

Any person may file a **post-grant opposition** within 12 months of grant (Section 25(2)). The patent holder must pay **annual renewal fees** (Form 4) to maintain the patent; non-payment results in lapsing of rights.

Timeline Summary: Priority Date → Publication (18 months) → Request for Examination (within 48 months) → FER Response (6 months) → Grant → Post-Grant Opposition window (12 months) → Annual Renewals (up to 20 years).

Conclusion: *The patent registration process in India is structured yet rigorous, designed to ensure only genuinely novel and inventive creations receive protection. Entrepreneurs and inventors should begin with a thorough prior-art search, file early to secure priority, and engage a registered Patent Agent for prosecution. The government's fast-track scheme for start-ups (expedited examination) and fee concessions for individuals further democratise access to the patent system.*

References

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6. Chesbrough, H. (2003). *Open Innovation: The New Imperative for Creating and Profiting from Technology.* Harvard Business School Press.

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8. National IPR Policy, 2016 — DPIIT, Ministry of Commerce & Industry.